

WEEK 1 – UNDERSTANDING THE BUSINESS PROBLEM

REQUIREMENT ANALYSIS DOCUMENT

PROJECT TITLE: E-COMMERCE SHOPPING PLATFORM

1. INTRODUCTION

The proposed system is an online shopping platform similar to Amazon or Flipkart. It allows customers to browse products, add products to a shopping cart, place orders, make payments, and track shipments.

The system provides a convenient platform for customers to purchase products online. It also allows administrators to manage products, customers, orders, payments, inventory, and deliveries.

2. BUSINESS OBJECTIVES

The main objectives of the proposed e-commerce system are:

- Allow customers to register and securely log in.
- Enable customers to browse and search for products.
- Display product details such as price, description, images, and availability.
- Allow customers to add products to a shopping cart.
- Enable customers to place orders.
- Provide secure payment processing.
- Allow customers to track their orders.
- Allow administrators to manage products and orders.
- Maintain customer, order, and transaction information efficiently.

3. MAJOR ACTIVITIES IN AN E-COMMERCE SYSTEM

The major activities involved in an online shopping system are:

Customer Registration



Product Browsing



Product Selection



Add to Cart



Order Placement



Payment Processing



Order Processing



Shipment

↓
Order Tracking

↓
Delivery

4. FUNCTIONAL REQUIREMENTS

4.1 CUSTOMER REGISTRATION AND LOGIN

The system should allow users to:

- Create a new account.
- Enter their name, email, phone number, and password.
- Log in using their registered credentials.
- Log out securely.
- Update their profile information.

4.2 PRODUCT BROWSING

Customers should be able to:

- View available products.
- Search for products.
- Filter products by category, price, and brand.
- Sort products based on price or popularity.
- View complete product details.
- Check product availability.

4.3 SHOPPING CART

The system should allow customers to:

- Add products to the cart.
- Remove products from the cart.
- Change product quantities.
- View the total cart amount.
- Continue shopping before placing an order.

4.4 ORDER PLACEMENT

Customers should be able to:

- Select products from their cart.
- Select or enter a delivery address.
- Review their order before confirmation.
- Confirm and place the order.
- Receive an order confirmation.

4.5 PAYMENT PROCESSING

The system should support different payment methods such as:

- Credit/Debit Card
- UPI
- Net Banking
- Cash on Delivery

The system should provide payment confirmation after a successful transaction.

4.6 SHIPMENT TRACKING

Customers should be able to:

- View their order status.
- Track their shipment.
- View estimated delivery information.

The order status can be:

Order Placed → Confirmed → Packed → Shipped → Out for Delivery → Delivered

4.7 ORDER MANAGEMENT

Customers should be able to:

- View their previous orders.
- Check individual order details.
- Cancel eligible orders.
- View the current order status.

5. ADMINISTRATOR REQUIREMENTS

The administrator should be able to:

- Log in to the admin panel.
- Add new products.
- Update product information.
- Remove products.
- Manage product categories.
- Manage customer accounts.
- View and manage orders.
- Update order status.
- Monitor payments.
- Manage product inventory.

6. NON-FUNCTIONAL REQUIREMENTS

SECURITY:

Customer and payment information must be protected.

PERFORMANCE:

The website should load pages and search results quickly.

USABILITY:

The system should have a simple and user-friendly interface.

SCALABILITY:

The system should support an increasing number of users and products.

AVAILABILITY:

The platform should be available whenever customers need it.

RELIABILITY:

Orders and payments should be processed accurately.

RESPONSIVENESS:

The website should work properly on mobile, tablet, and desktop devices.

7. MAIN ACTORS

CUSTOMER:

- Register and login
- Browse products
- Search products
- Add products to cart
- Place orders
- Make payments
- Track shipments
- View order history

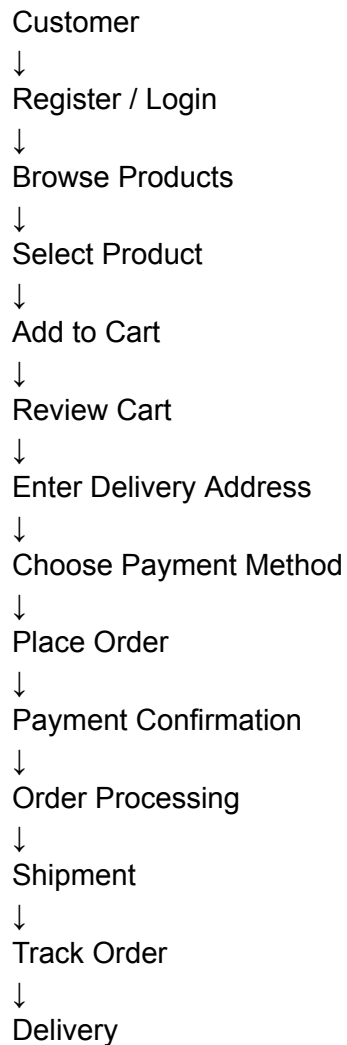
ADMINISTRATOR:

- Manage products
- Manage customers
- Manage inventory
- Manage orders
- Update order status
- Monitor payments

DELIVERY/SHIPPING SYSTEM:

- Receive shipment information
- Process delivery
- Update delivery status
- Provide tracking information

8. SYSTEM WORKFLOW



9. PROPOSED SYSTEM MODULES

The proposed e-commerce platform can be divided into the following modules:

1. User Management
2. Product Management
3. Category Management
4. Shopping Cart Management
5. Order Management
6. Payment Management
7. Shipment and Tracking
8. Inventory Management
9. Admin Management
10. CONCLUSION

The proposed e-commerce system provides customers with a complete online shopping experience, starting from registration and product browsing to payment, shipment tracking, and delivery.

The requirement analysis identifies the major functionalities and business activities required to develop the system. This document can be used as the foundation for the next stages of the project, including system design, database design, user interface development, and implementation.